

Ghulam Ishaq Khan Institute (GIKI)

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

ASSIGNMENT #3

Course Information	
Subject	Database Management System
Course Code	CS-232-A-Spring-26
Class	BSAI – 4th Semester, Batch 34 (Fall-24 Intake)
Total Marks	10
Instructor	M. Qasim Riaz
Submission Deadline	10/May/2026 – Sunday (11:59 PM)

Note (Read notes & instructions of each exercise first):

- Will be submitted by all members of the group (Will be marked for each student who submitted).
- Each group member who submits assignment will write his/her name & reg# first in Section 2 where names and reg numbers of group members are being asked to be added (this will be considered as part of grading).

Project Report Component – 3 (LIST OF TABLES + ERD + SCHEMA OF DATABASE)

Hospify — Hospital Management System

1. Group Members

#	Name	Registration #
1	Atta Ur Rahman Sheikh	2024122
2	Ibrahim Umar	2024413
3	Talal Iqbal	2024476

2. List of All Tables

The Hospify database consists of **21 tables** in 3NF, listed below in foreign-key-safe creation order.

#	Table Name	Description
1	users	System accounts for all roles (admin, doctor, nurse, etc.)
2	patients	Patient demographics, CNIC, blood group, emergency contact
3	wards	Hospital wards with type, floor, and bed capacity
4	beds	Individual beds linked to wards with status tracking
5	doctors	Doctor profiles linked to user accounts
6	nurses	Nurse profiles linked to user accounts and wards
7	lab_technicians	Lab technician profiles linked to user accounts
8	pharmacists	Pharmacist profiles linked to user accounts
9	admissions	Patient admissions with bed, doctor, and status info
10	appointments	Scheduled appointments between patients and doctors
11	vitals	Patient vitals recorded by nurses during admissions
12	medicines	Medicine catalogue with generic/brand names and category
13	medicine_inventory	Stock levels and reorder thresholds per medicine
14	prescriptions	Prescription headers linked to admissions and doctors
15	prescription_items	Line items per prescription (medicine, dosage, frequency)
16	media_files	Binary file storage for lab results (BYTEA + Firebase URL)
17	lab_orders	Lab test orders placed by doctors for admitted patients
18	lab_results	Results uploaded by lab technicians for completed orders
19	bills	Bill headers generated from admissions
20	bill_items	Itemised bill lines (bed days, labs, prescriptions)
21	payments	Payment records against bills
22	alerts	Low-stock and abnormal-vitals alert records
23	audit_log	Full JSONB before/after snapshots of all table changes

3. Entity-Relationship Diagram (ERD) & Schema Diagram

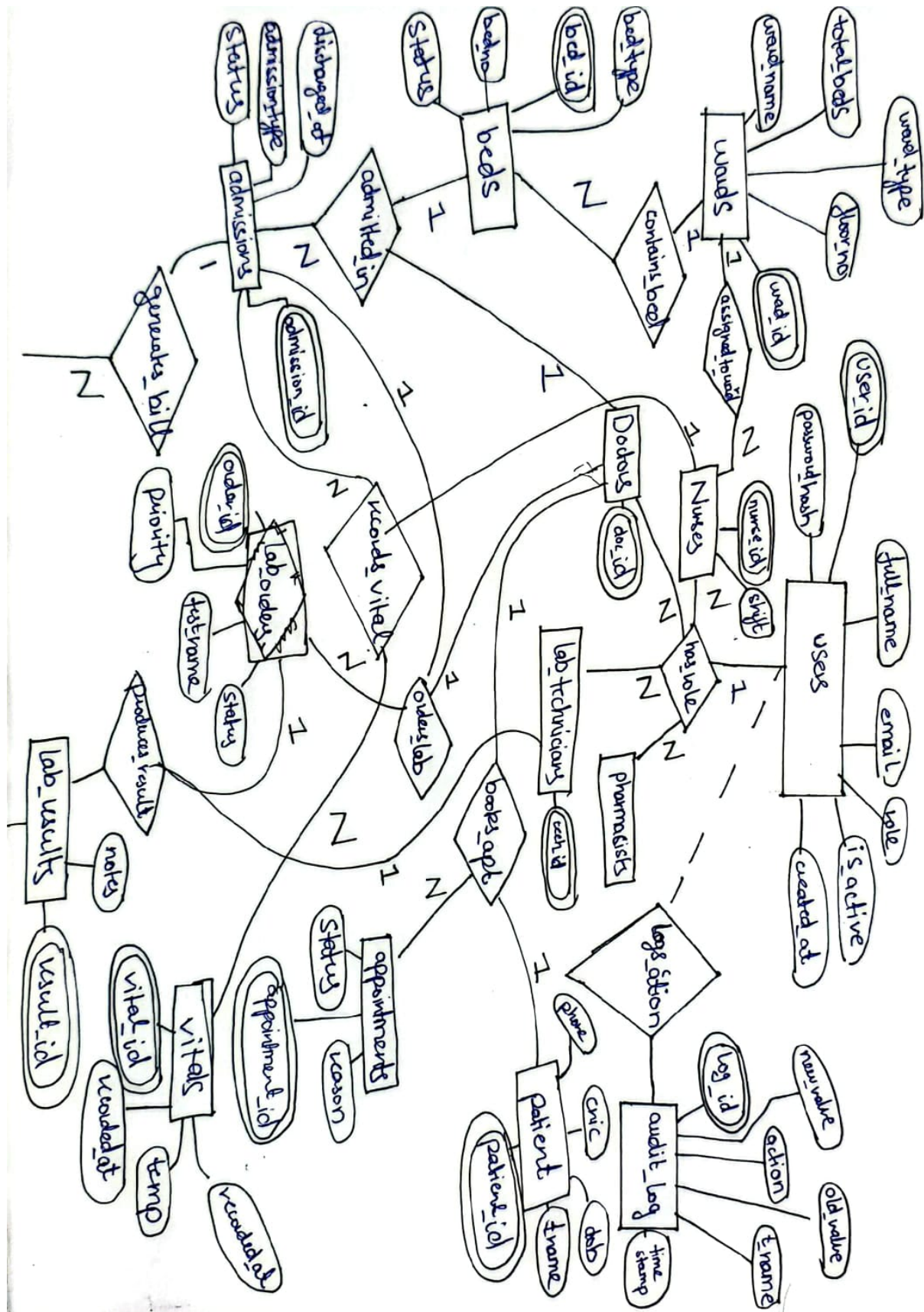


Figure 1: Enter Caption

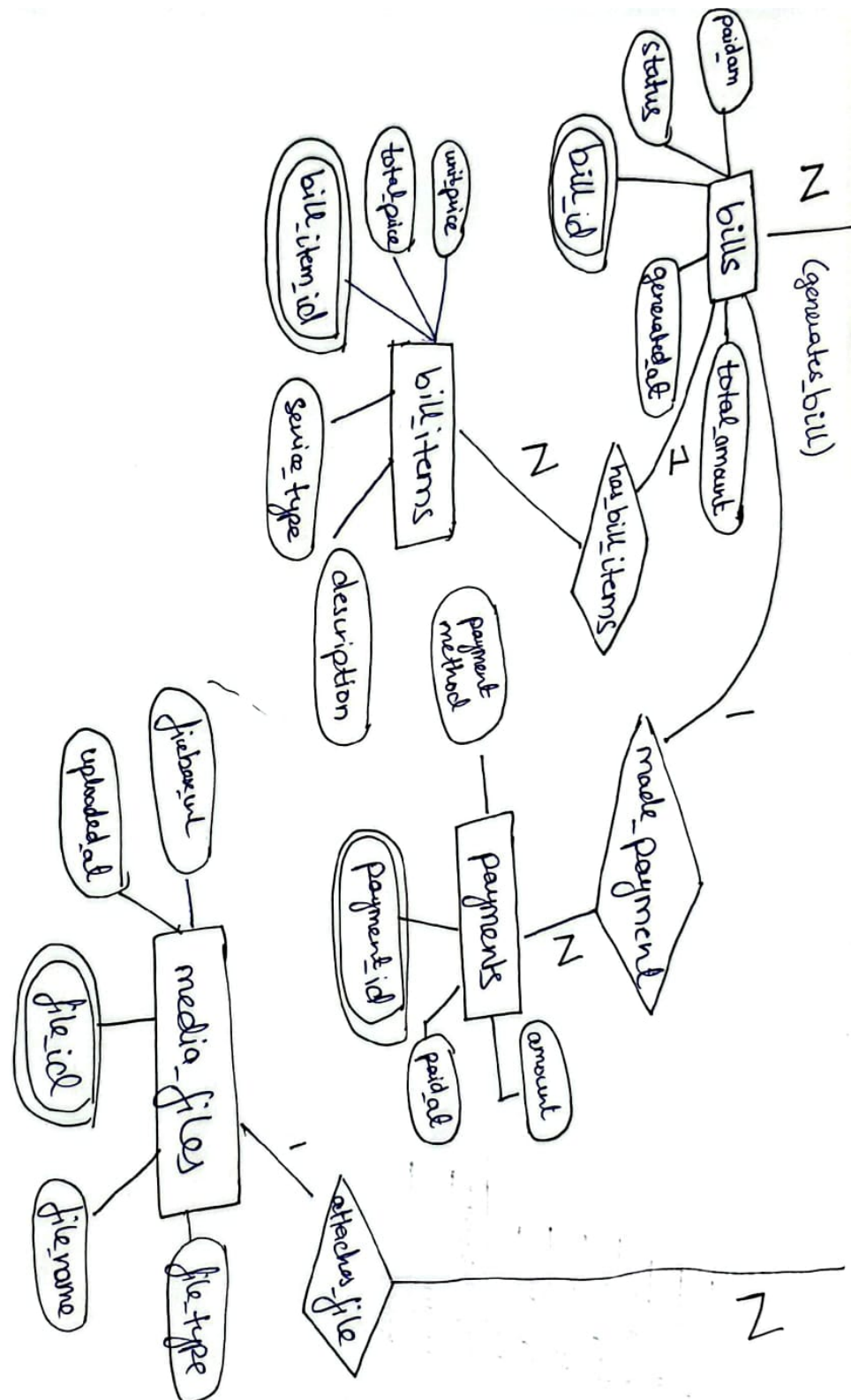


Figure 2: Enter Caption

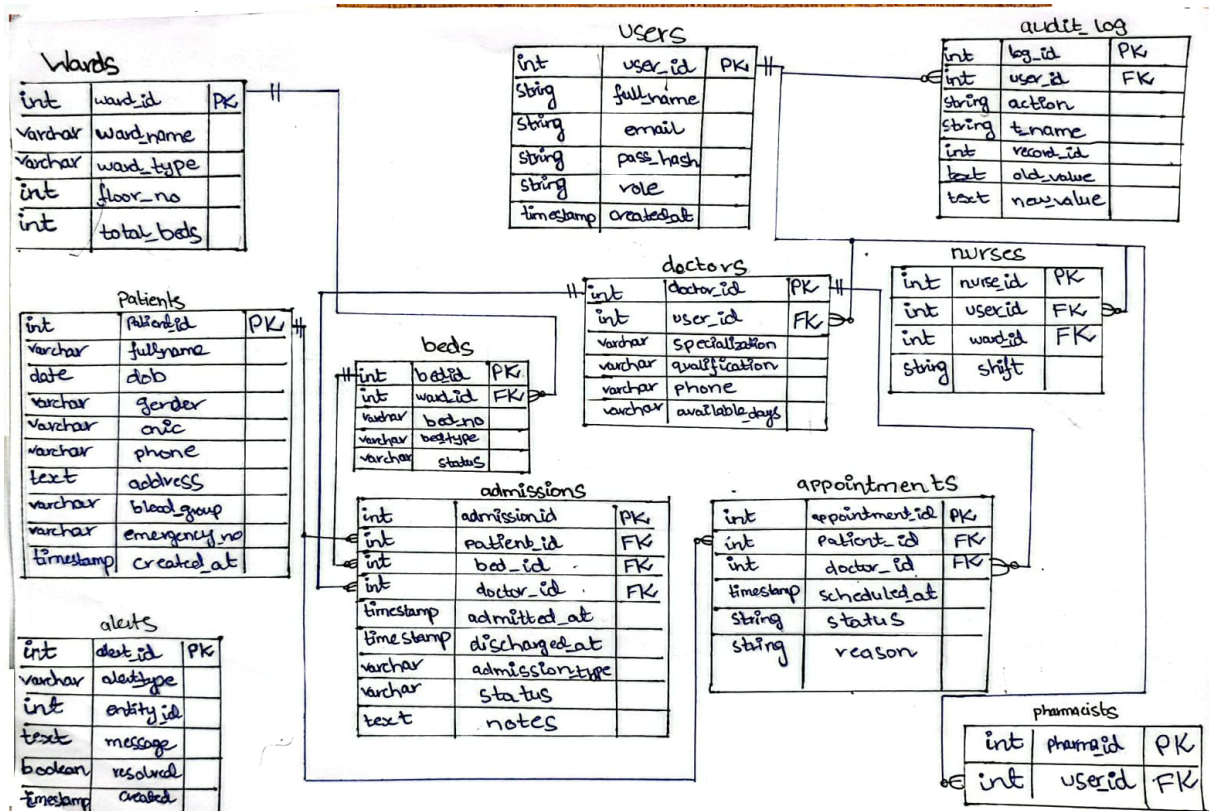


Figure 3: Enter Caption

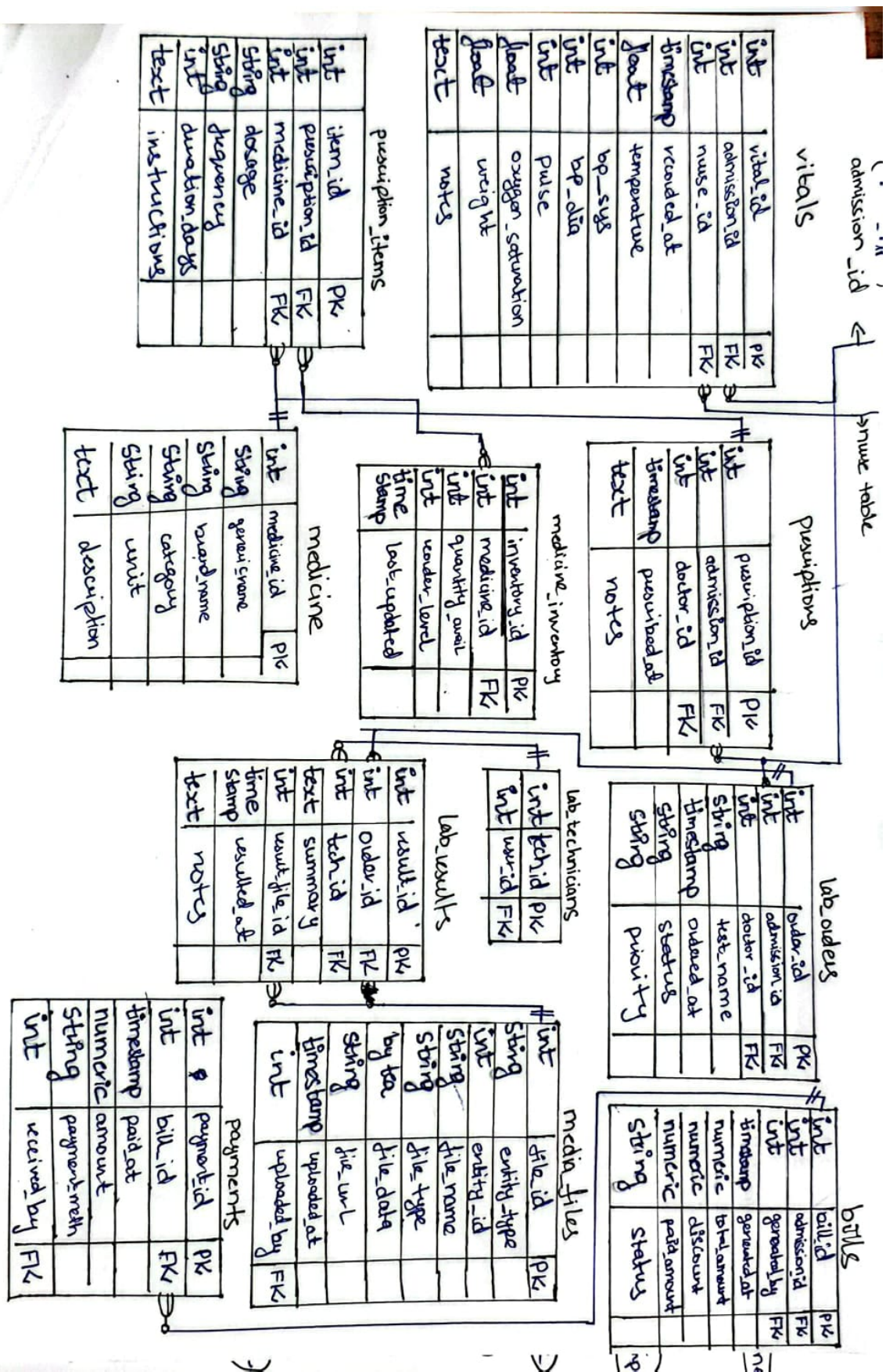


Figure 4: Enter Caption